

Data dictionary

Client: One row represents one client (a buyer or seller) registered with TotallyRealEstate.

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
client_id	Unique identifier	Integer	PK	NOT NULL	1001
first_name	First name of client	Varchar		NOT NULL	John
last_name	Last name of client	Varchar		NOT NULL	Doe
email	Contact email	Varchar	UK	Optional; UNIQUE	Johndoe@gmail.com
phone	Contact phone	varchar		NOT NULL	403-123-4567
created_at	When client account was created	timestampz		NOT NULL	2026-10-06 10:00-06:00

Agent: One row represents an independent real estate agent affiliated with TotallyRealEstate.

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
agent_id	Unique identifier	Integer	PK	NOT NULL	1001
license_number	Agents' real estate licence	Varchar	UK	NOT NULL; UNIQUE	RE-ag-123
first_name	first name of agent	Varchar		NOT NULL	Agent
last_name	Last name of agent	Varchar		NOT NULL	bob
email	Contact email	varchar	UK	NOT NULL; UNIQUE	Agentbob@gmail.com
phone	Contact phone	varchar		NOT NULL	403-123-4567
is_active	Is the agent active	boolean		NOT NULL	true

Staff: One row represents an internal, full-time administrative or operations employee at TotallyRealEstate.

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
staff_id	Unique identifier	Integer	PK	NOT NULL	1001
first_name	Staff's first name	Varchar		NOT NULL	Staff
last_name	Staff's last name	Varchar		NOT NULL	member
email	Contact email	Varchar	UK	NOT NULL; UNIQUE	Staffmember@gmail.com
role_code	Code that specifies role	Varchar		NOT NULL; see roles below	COORDINATOR
is_active	Current employment status	boolean		NOT NULL	true

role_code: BROKER, ADMIN, COORDINATOR, ACCOUNTANT or MARKETING.

Property: One row represents a unique real estate asset managed in the system

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
property_id	Unique identifier	integer	PK	NOT NULL	1001
address_text	Property street address	Varchar		NOT NULL	67 tickle boulevard
property_ref	Municipal property code	Varchar	UK	NOT NULL; UNIQUE	A-123
municipality	City where property is located	Varchar		NOT NULL	Calgary
property_type	Type of property	Varchar		NOT NULL; see types below	CONDO
bedrooms	Amount of bedrooms	integer		Optional; integer >= 0	2
bathrooms	Amount of bathrooms	numeric		Optional; >= 0	2
floor_area_sqm	Total square metres in property	numeric		Optional; > 0	92.90

property_type: DETACHED, CONDO, TOWNHOUSE or OTHER.

Listing: One row represents an individual market listing for a property

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
listing_id	Unique identifier	integer	PK	NOT NULL	1001
property_id	Foreign key referencing the property	integer	FK	NOT NULL	1234
agent_id	Foreign key referencing the agent	integer	FK	NOT NULL	6769
asking_price	Initial or current listed sale price	numeric		NOT NULL; > 0	650,000
listed_on	Date the listing went on the market	date		NOT NULL	2026-10-06
closed_on	Date the listing was sold or withdrawn	date		Optional; >= listed_on	2026-11-06
status	Status of listing	varchar		NOT NULL; see states below	SOLD

status: DRAFT, ACTIVE, UNDER_OFFER, SOLD, WITHDRAWN or EXPIRED. DRAFT, ACTIVE and UNDER_OFFER have NULL closed_on. SOLD, WITHDRAWN and EXPIRED require closed_on. Each property has at most one ACTIVE or UNDER_OFFER listing. Non-draft listings require at least one seller. Foreign keys: property_id -> PROPERTY.property_id; agent_id -> AGENT.agent_id.

Listing seller: One row represents an associative link attaching one seller client as an owner of a property listing

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
listing_id	Foreign key identifying listing	integer	PK/FK	NOT NULL	1001
client_id	Foreign key identifying client	integer	PK/FK	NOT NULL	1234

Composite primary key: (listing_id, client_id). Foreign keys: listing_id -> LISTING.listing_id; client_id -> CLIENT.client_id.

Showing: One row represents a scheduled viewing appointment time slot hosted by an agent for a property listing

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
showing_id	Unique identifier	Integer	PK	NOT NULL	1001
listing_id	Foreign key identifying listing	Integer	FK	NOT NULL	1234
agent_id	Foreign key identifying agent	Integer	FK	NOT NULL	6769
starts_at	Date and time the showing begins	timestampz		NOT NULL	2026-10-06 14:00-06:00
ends_at	Date and time the showing ends	timestampz		NOT NULL; > starts_at	2026-11-06 14:30-07:00
status	Outcome of showing	varchar		NOT NULL; see states below	SCHEDULED

status: SCHEDULED, COMPLETED or CANCELLED. Non-cancelled showing intervals [starts_at, ends_at) must not overlap for the same property or agent. Foreign keys: listing_id -> LISTING.listing_id; agent_id -> AGENT.agent_id.

Showing Booking: One row represents a client's specific booking for a property showing appointment

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
booking_id	Unique identifier	integer	PK	NOT NULL	1001
showing_id	Foreign key identifying showing	Integer	FK	NOT NULL	1234
client_id	Foreign key identifying lead client	Integer	FK	NOT NULL	6769
party_size	Amount of people viewing property	integer		NOT NULL; 1 to 3; includes lead client	3
status	Outcome of booking	varchar		NOT NULL; see states below	CONFIRMED
booked_at	Time the showing is booked at	timestampz		NOT NULL	2026-10-10 10:00-06:00

status: PENDING, CONFIRMED, CANCELLED or REJECTED. PENDING and CONFIRMED bookings are active. Each showing allows at most 3 active bookings; each showing/client pair has at most 1 active booking. Foreign keys: showing_id -> SHOWING.showing_id; client_id -> CLIENT.client_id.

Offer: One row represents a formal purchase proposal submitted by an agent for a property listing

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
offer_id	Unique identifier	integer	PK	NOT NULL	1001
listing_id	Foreign key identifying listing	integer	FK	NOT NULL	1234
submitted_by_agent_id	Foreign key identifying agent	integer	FK	NOT NULL	6768
offer_amount	Proposed purchase amount	numeric		NOT NULL; > 0	650,000
submitted_at	Timestamp when offer was sent	timestamptz		NOT NULL	2026-10-06 10:00-06:00
expires_at	Timestamp when offer expires	timestamptz		Optional; > submitted_at	2026-11-06 18:00-07:00
status	Current processing status of the offer	Varchar		NOT NULL; see states below	ACCEPTED
accepted_at	Timestamp when the offer was accepted	timestamptz		Only for ACCEPTED; see dates below	2026-10-21 10:00-06:00

status: SUBMITTED, ACCEPTED, REJECTED, WITHDRAWN or EXPIRED. Each offer requires at least one buyer. Each listing allows at most one ACCEPTED offer. All other offer states have NULL accepted_at. accepted_at must be >= submitted_at and <= expires_at when an expiry is present. Foreign keys: listing_id -> LISTING.listing_id; submitted_by_agent_id -> AGENT.agent_id.

Offer buyer: One row represents an associative link attaching one buyer client to a submitted purchase offer

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
offer_id	Foreign key identifying offer	integer	PK/FK	NOT NULL	1001
client_id	foreign key identifying client	integer	PK/FK	NOT NULL	1234

Composite primary key: (offer_id, client_id). Foreign keys: offer_id -> OFFER.offer_id; client_id -> CLIENT.client_id.

Sale transaction: One row represents a completed real estate sale finalized from an accepted purchase offer

Attribute	Description	Data type/Format	Key	Validation rules	Sample/Default
sale_id	Unique identifier	Integer	PK	NOT NULL	1001
offer_id	Foreign key identifying offer	integer	FK/UK	NOT NULL; UNIQUE	1234
coordinator_staff_id	Foreign key identifying staff	integer	FK	NOT NULL	6769
completed_on	official date of transaction	date		NOT NULL; not before offer acceptance date	2026-11-06
sale_price	Final price of the sale	numeric		NOT NULL; > 0	650,000

The offer must be ACCEPTED. The coordinator must be active STAFF with role_code COORDINATOR. Completion uses the Calgary-local acceptance date. All foreign keys use ON DELETE RESTRICT and ON UPDATE RESTRICT. Cancelled and closed records are retained. Foreign keys: offer_id -> OFFER.offer_id; coordinator_staff_id -> STAFF.staff_id.